

SHETH L.U.J AND SIR M.V. COLLEGE
SUBJECT NAME: Cyber & Information Security
Practical 5 :Key Exchange usingDiffie-Hellman

Aim : Implement the Diffie-Hellman key exchange algorithm to securely exchange keys between two entities over an insecure network.

Code :

```
5p.py - C:/Users/IT-49/AppData/Local/Programs/Python/Python314/5p.py (3.14.6)
File Edit Format Run Options Window Help
p = 23
g = 5

alice_private = 6
bob_private = 15

alice_public = pow(g, alice_private, p)
bob_public = pow(g, bob_private, p)

print("Alice's public key:", alice_public)
print("Bob's public key:", bob_public)

alice_shared_secret = pow(bob_public, alice_private, p)
bob_shared_secret = pow(alice_public, bob_private, p)

print("Alice's shared secret:", alice_shared_secret)
print("Bob's shared secret:", bob_shared_secret)

if alice_shared_secret == bob_shared_secret:
    print("Key exchange successful!")
    print("Shared secret:", alice_shared_secret)
else:
    print("Key exchange failed!")
```

Output :

```
==== RESTART: C:/Users/IT-49/AppData/Local/Programs/Python/Python314/5p.py ====
Alice's public key: 8
Bob's public key: 19
Alice's shared secret: 2
Bob's shared secret: 2
Key exchange successful!
Shared secret: 2
>>>
```

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#2 Implement the Diffie-Hellman key exchange algorithm to securely exchange keys between two entities over an insecure network User input.

```
5p user input.py - C:/Users/IT-49/AppData/Local/Programs/Python/Python314/5p user input.py (3.14.6)
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p = int(input("Enter a prime number (p): "))
g = int(input("Enter a primitive root (g): "))

ram_private = int(input("Enter Ram's private key: "))
sachine_private = int(input("Enter Sachine's private key: "))

ram_public = pow(g, ram_private, p)
sachine_public = pow(g, sachine_private, p)

print("\n--- Public Key Exchange ---")
print("Ram's public key:", ram_public)
print("Sachine's public key:", sachine_public)

ram_shared_secret = pow(sachine_public, ram_private, p)
sachine_shared_secret = pow(ram_public, sachine_private, p)

print("\n--- Shared Secret ---")
print("Ram's shared secret:", ram_shared_secret)
print("Sachine's shared secret:", sachine_shared_secret)

if ram_shared_secret == sachine_shared_secret:
    print("\nKey exchange successful!")
    print("Shared secret key:", ram_shared_secret)
else:
```

Output :

```
>>>
===== RESTART: C:/Users/IT-49/AppData/Local/Programs/Python/Python314/5p user input.py =====
Enter a prime number (p): 11
Enter a primitive root (g): 117
Enter Ram's private key: 255
Enter Sachine's private key: 333

--- Public Key Exchange ---
Ram's public key: 10
Sachine's public key: 2

--- Shared Secret ---
Ram's shared secret: 10
Sachine's shared secret: 10

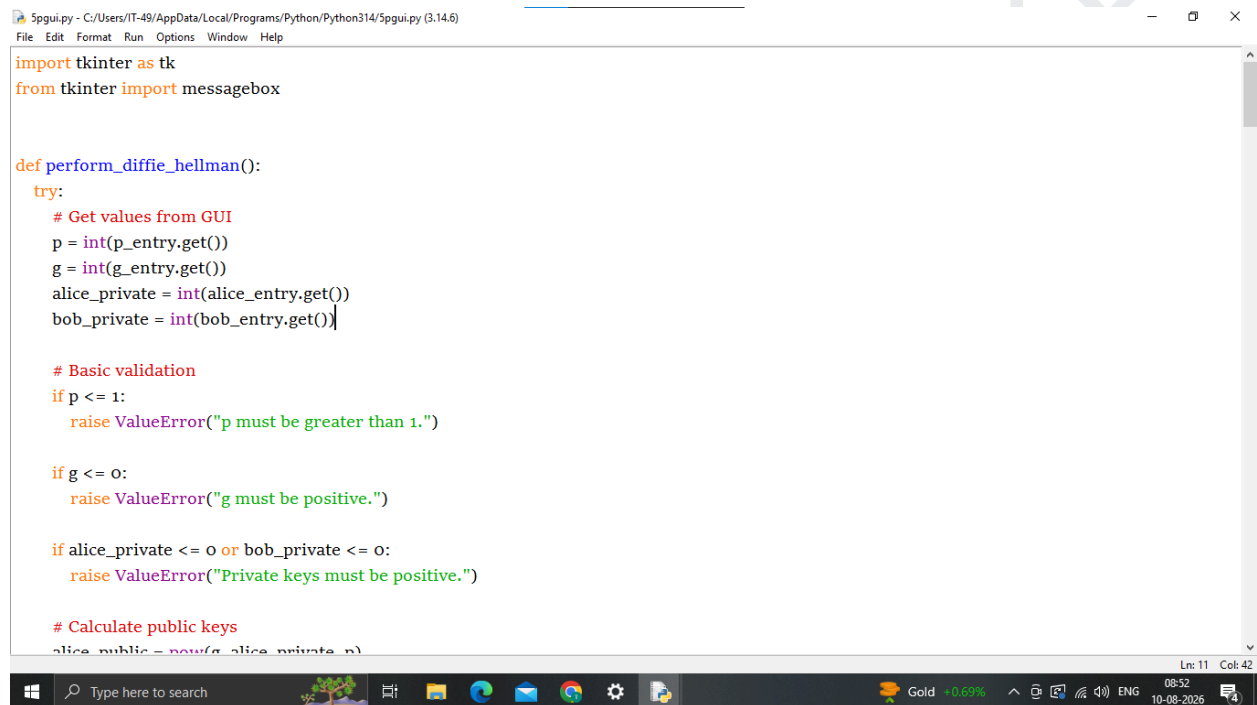
Key exchange successful!
Shared secret key: 10
>>>
```

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Implement the Diffie-Hellman key exchange algorithm to securely exchange keys between two entities over an insecure network. Gui



```
5pgui.py - C:/Users/IT-49/AppData/Local/Programs/Python/Python314/5pgui.py (3.14.6)
File Edit Format Run Options Window Help

import tkinter as tk
from tkinter import messagebox

def perform_diffie_hellman():
    try:
        # Get values from GUI
        p = int(p_entry.get())
        g = int(g_entry.get())
        alice_private = int(alice_entry.get())
        bob_private = int(bob_entry.get())

        # Basic validation
        if p <= 1:
            raise ValueError("p must be greater than 1.")

        if g <= 0:
            raise ValueError("g must be positive.")

        if alice_private <= 0 or bob_private <= 0:
            raise ValueError("Private keys must be positive.")

        # Calculate public keys
        alice_public = pow(g, alice_private, p)
```

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```
Spgui.py - C:/Users/IT-49/AppData/Local/Programs/Python/Python314/Spgui.py (3.14.6)
File Edit Format Run Options Window Help

try:
    # Get values from GUI
    p = int(p_entry.get())
    g = int(g_entry.get())
    alice_private = int(alice_entry.get())
    bob_private = int(bob_entry.get())

    # Basic validation
    if p <= 1:
        raise ValueError("p must be greater than 1.")

    if g <= 0:
        raise ValueError("g must be positive.")

    if alice_private <= 0 or bob_private <= 0:
        raise ValueError("Private keys must be positive.")

    # Calculate public keys
    alice_public = pow(g, alice_private, p)
    bob_public = pow(g, bob_private, p)

    # Calculate shared secret
    alice_shared = pow(bob_public, alice_private, p)
    bob_shared = pow(alice_public, bob_private, p)

Ln: 11 Col: 42
```

```
*Spgui.py - C:/Users/IT-49/AppData/Local/Programs/Python/Python314/Spgui.py (3.14.6)*
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    raise ValueError("g must be positive.")

if alice_private <= 0 or bob_private <= 0:
    raise ValueError("Private keys must be positive.")

alice_public = pow(g, alice_private, p)
bob_public = pow(g, bob_private, p)

alice_shared = pow(bob_public, alice_private, p)
bob_shared = pow(alice_public, bob_private, p)

alice_public_result.config(text=str(alice_public))
bob_public_result.config(text=str(bob_public))
alice_shared_result.config(text=str(alice_shared))
bob_shared_result.config(text=str(bob_shared))

if alice_shared == bob_shared:
    status_label.config(
        text="✓ Key Exchange Successful!",
        fg="green"
    )
else:
    status_label.config(
        text="✗ Key Exchange Failed!"

Ln: 27 Col: 0
```

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```
*Spgui.py - C:/Users/IT-49/AppData/Local/Programs/Python/Python314/Spgui.py (3.14.6)*
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def clear_fields():
    p_entry.delete(0, tk.END)
    g_entry.delete(0, tk.END)
    alice_entry.delete(0, tk.END)
    bob_entry.delete(0, tk.END)

    alice_public_result.config(text="-")
    bob_public_result.config(text="-")
    alice_shared_result.config(text="-")
    bob_shared_result.config(text="-")

    status_label.config(text="")

root = tk.Tk()
root.title("Diffie-Hellman Key Exchange")
root.geometry("600x600")
root.resizable(False, False)
root.configure(bg="#f2f2f2")

title = tk.Label(
    root,
    text="Diffie-Hellman Key Exchange"
```

```
*Spgui.py - C:/Users/IT-49/AppData/Local/Programs/Python/Python314/Spgui.py (3.14.6)*
File Edit Format Run Options Window Help

title = tk.Label(
    root,
    text="Diffie-Hellman Key Exchange",
    font=("Arial", 22, "bold"),
    bg="#f2f2f2",
    fg="#1f3c88"
)
title.pack(pady=20)

subtitle = tk.Label(
    root,
    text="Secure key exchange demonstration",
    font=("Arial", 11),
    bg="#f2f2f2",
    fg="#555555"
)
subtitle.pack()

input_frame = tk.Frame(root, bg="white", padx=20, pady=20)
input_frame.pack(pady=20)

SStk.Label(
    input_frame
```

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```
*5pgui.py - C:/Users/IT-49/AppData/Local/Programs/Python/Python314/5pgui.py (3.14.6)*
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input_frame.pack(pady=20)

tk.Label(
    input_frame,
    text="Public Parameters",
    font=("Arial", 14, "bold"),
    bg="white"
).grid(row=0, column=0, columnspan=2, pady=10)

tk.Label(
    input_frame,
    text="Prime number (p):",
    font=("Arial", 11),
    bg="white"
).grid(row=1, column=0, sticky="w", pady=7)

_entry = tk.Entry(input_frame, width=25, font=("Arial", 11))
_entry.grid(row=1, column=1, pady=7)

tk.Label(
    input_frame,
    text="Generator (g):",
    font=("Arial", 11),
    bg="white"
).grid(row=2, column=0, sticky="w", pady=7)

g_entry = tk.Entry(input_frame, width=25, font=("Arial", 11))
g_entry.grid(row=2, column=1, pady=7)

tk.Label(
    input_frame,
    text="Private Keys",
    font=("Arial", 14, "bold"),
    bg="white"
).grid(row=3, column=0, columnspan=2, pady=(15, 10))

tk.Label(
    input_frame,
    text="Alice's private key:",
    font=("Arial", 11),
    bg="white"
).grid(row=4, column=0, sticky="w", pady=7)

alice_entry = tk.Entry(input_frame, width=25, font=("Arial", 11))
alice_entry.grid(row=4, column=1, pady=7)

tk.Label(
    input_frame,
    text="Bob's private key:"

```

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```
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File Edit Format Run Options Window Help

button_frame = tk.Frame(root, bg="#f2f2f2")
button_frame.pack(pady=5)

calculate_button = tk.Button(
    button_frame,
    text="Calculate Shared Key",
    command=perform_diffie_hellman,
    bg="#1f6feb",
    fg="white",
    font=("Arial", 11, "bold"),
    padx=15,
    pady=8
)
calculate_button.grid(row=0, column=0, padx=10)

clear_button = tk.Button(
    button_frame,
    text="Clear",
    command=clear_fields,
    bg="#555555",
    fg="white",
    font=("Arial", 11, "bold"),
    padx=15,
    pady=8
)
clear_button.grid(row=1, column=0, padx=10)

Ln: 142 Col: 0
```

```
*Spgui.py - C:/Users/IT-49/AppData/Local/Programs/Python/Python314/Spgui.py (3.14.6)*
File Edit Format Run Options Window Help

    bg="white"
).grid(row=0, column=0, columnspan=2, pady=5)

tk.Label(
    result_frame,
    text="Alice's Public Key:",
    font=("Arial", 11),
    bg="white"
).grid(row=1, column=0, sticky="w", pady=5)

alice_public_result = tk.Label(
    result_frame,
    text="",
    font=("Arial", 11, "bold"),
    bg="white",
    fg="#1f3c88"
)
alice_public_result.grid(row=1, column=1, sticky="w", padx=20)

tk.Label(
    result_frame,
    text="Bob's Public Key:",
    font=("Arial", 11),
    bg="white"
).grid(row=2, column=0, sticky="w", pady=5)

Ln: 169 Col: 0
```

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```
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File Edit Format Run Options Window Help
text="Bob's Shared Secret:",
font=("Arial", 11),
bg="white"
).grid(row=4, column=0, sticky="w", pady=5)

bob_shared_result = tk.Label(
    result_frame,
    text="-",
    font=("Arial", 11, "bold"),
    bg="white",
    fg="#8a2be2"
)
bob_shared_result.grid(row=4, column=1, sticky="w", padx=20)

status_label = tk.Label(
    root,
    text="",
    font=("Arial", 13, "bold"),
    bg="#f2f2f2"
)
status_label.pack(pady=10)

root.mainloop()
```

Output

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Diffie-Hellman Key Exchange

Diffie-Hellman Key Exchange

Secure key exchange demonstration

Public Parameters

Prime number (p):

Generator (g):

Private Keys

Alice's private key:

Bob's private key:

Results

Alice's Public Key: **8**

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